In [1]:
```
import pandas as pd
```

In [2]:
```
dataset = pd.read_csv('Salary_dataset.csv')
```

In [3]:
```
dataset
```

Out[3]:

	Unnamed: 0	YearsExperience	Salary
0	0	1.2	39344.0
1	1	1.4	46206.0
2	2	1.6	37732.0
3	3	2.1	43526.0
4	4	2.3	39892.0
5	5	3.0	56643.0
6	6	3.1	60151.0
7	7	3.3	54446.0
8	8	3.3	64446.0
9	9	3.8	57190.0
10	10	4.0	63219.0
11	11	4.1	55795.0
12	12	4.1	56958.0
13	13	4.2	57082.0
14	14	4.6	61112.0
15	15	5.0	67939.0
16	16	5.2	66030.0
17	17	5.4	83089.0
18	18	6.0	81364.0
19	19	6.1	93941.0
20	20	6.9	91739.0
21	21	7.2	98274.0
22	22	8.0	101303.0
23	23	8.3	113813.0
24	24	8.8	109432.0
25	25	9.1	105583.0
26	26	9.6	116970.0
27	27	9.7	112636.0
28	28	10.4	122392.0
29	29	10.6	121873.0

In [4]:
```
dataset = dataset.drop(columns=['Unnamed: 0'])
```

In [5]:
```
dataset
```

Out[5]:

	YearsExperience	Salary
0	1.2	39344.0
1	1.4	46206.0
2	1.6	37732.0
3	2.1	43526.0
4	2.3	39892.0
5	3.0	56643.0
6	3.1	60151.0
7	3.3	54446.0
8	3.3	64446.0
9	3.8	57190.0
10	4.0	63219.0
11	4.1	55795.0
12	4.1	56958.0
13	4.2	57082.0
14	4.6	61112.0
15	5.0	67939.0
16	5.2	66030.0
17	5.4	83089.0
18	6.0	81364.0
19	6.1	93941.0
20	6.9	91739.0
21	7.2	98274.0
22	8.0	101303.0
23	8.3	113813.0
24	8.8	109432.0
25	9.1	105583.0
26	9.6	116970.0
27	9.7	112636.0
28	10.4	122392.0
29	10.6	121873.0

In [6]:
```
dataset.isnull().sum()
```

Out[6]:
```
YearsExperience    0
Salary             0
dtype: int64
```

In [7]:
```
x = dataset[["YearsExperience"]]
y = dataset[["Salary"]]
```

In [8]:
```
y
```

Out[8]:

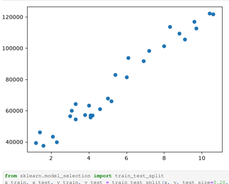
	Salary
0	39344.0
1	46206.0
2	37732.0
3	43526.0
4	39892.0
5	56643.0
6	60151.0
7	54446.0
8	64446.0
9	57190.0
10	63219.0
11	55795.0
12	56958.0
13	57082.0
14	61112.0
15	67939.0
16	66030.0
17	83089.0
18	81364.0
19	93941.0
20	91739.0
21	98274.0
22	101303.0
23	113813.0
24	109432.0
25	105583.0
26	116970.0
27	112636.0
28	122392.0
29	121873.0

In [9]:
```
x
```

Out[9]:

	YearsExperience
0	1.2
1	1.4
2	1.6
3	2.1
4	2.3
5	3.0
6	3.1
7	3.3
8	3.3
9	3.8
10	4.0
11	4.1
12	4.1
13	4.2
14	4.6
15	5.0
16	5.2
17	5.4
18	6.0
19	6.1
20	6.9
21	7.2
22	8.0
23	8.3
24	8.8
25	9.1
26	9.6
27	9.7
28	10.4
29	10.6

In [10]:
```
import matplotlib.pyplot as plt
plt.scatter(x,y)
plt.show()
```



In [11]:
```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.20, random_state=3)
```

In [12]:
```
y_test
```

Out[12]:

	Salary
9	57190.0
7	54446.0
0	39344.0
5	56643.0
6	60151.0
28	122392.0

In [13]:
```
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(x_train, y_train)
```

Out[13]:
```
LinearRegression
▸ Parameters
▸ Fitted attributes
```

In [14]:
```
y_pred = regressor.predict(x_test)
y_pred
```

Out[14]:
```
array([[ 60112.26842563],
       [ 55292.66112199],
       [ 35050.31044483],
       [ 32450.89673977],
       [ 53364.83820051],
       [123731.08483454]])
```

In [15]:
```
from sklearn.metrics import r2_score
result_accuracy = r2_score(y_test, y_pred)
result_accuracy
```

Out[15]:
```
0.9778242092590167
```

In [16]:
```
salary = regressor.predict([[1.4]])
salary
```

```
C:\Users\Admin\anaconda3\lib\site-packages\sklearn\utils\validation.py:2927: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

Out[16]:
```
array([[36979.15393679]])
```

In [17]:
```
salary = regressor.predict([[15]])
salary
```

```
C:\Users\Admin\anaconda3\lib\site-packages\sklearn\utils\validation.py:2927: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

Out[17]:
```
array([[148071.47202859]])
```

In [18]:
```
m = regressor.coef_
m
```

Out[18]:
```
array([[9639.2146078]])
```

In [19]:
```
import pickle
filename = "LINR_MS_MODEL.sav"
```

```
pickle.dump(regressor, open(filename, "wb"))
```

In [ ]:

```python
import streamlit as st
import pickle
import numpy as np

model = pickle.load(open("11AM_ML_MODEL.sav","rb"))
st.set_page_config()

st.title("Salary Prediction App")
st.write("Predict an employee's salary based on their Exp")
st.divider()
years_exp = st.number_input(
    "Enter the exp",
    min_value = 0.0,
    max_value = 50.0,
    value = 1.0,
    step = 0.1
)
if st.button("Predict Salary"):
    input_data = np.array([[years_exp]])
    prediction = model.predict(input_data)
    predict_salary = prediction[0][0]
    st.success(
        f"Predicted Salary: {predict_salary:,.2f}"
    )
```

Salary Prediction App

Predict an employee's salary based on their Exp

Enter the exp

15.00 − +

Predict Salary

Predicted Salary: 168,071.47